

HOLIDAY ASSIGNMENT

CLASS 8 MATHEMATICS



COMPLETE CLASS EIGHT HOME ASSIGNMENT



MATHEMATICS CLASS 8

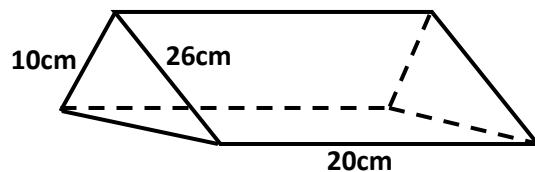
1. What is the value of a in the equation?

$$\frac{3a - 5}{4} + \frac{2a + 3}{3} = 4$$

2. In a school prize giving meeting, there were 240 women and half as many men. There were twice as many children as adults. How many people were in the meeting altogether?
3. In a polling station, there were 900 registered voters. During an election 10% of the registered voters did not cast their votes. The rest cast their votes to Ahmed, Babu and Chambuko in the ratio 3:2:1. If 30 votes were spoilt, how many votes did Ahmed get?
4. A cylindrical container has a capacity of 2.31litres. If the diameter of the container is 14cm, what is its height in cm? Take ($\pi = \frac{22}{7}$)
5. Simiyu bought 3 packets of sweets each containing 120 sweets at Sh. 50 per packet. He sold all the sweets at Sh. 2 for every 3 sweets. What percentage profit did he make?
6. A rectangular tank measures 1.2m by 80cm by 50cm high. Water is poured into the tank to a height of 15cm. How many more litres of water are needed to fill the tank?
7. Four boys contributed some money to buy a football. Musa contributed $\frac{1}{5}$, Simiyu $\frac{1}{6}$, Musyoka $\frac{1}{3}$ and Jabir $\frac{1}{8}$ of the price of the football. They still needed Sh. 84 to buy the football. What was the price of the ball?



8. A square plot of land of side 80m is represented on a scale drawing by a square whose perimeter is 16cm. what is the scale used?
9. Mwara is paid a basic salary of Sh. 7,500 per month. She is also paid a commission of $3\frac{1}{2}\%$ on all sales above Sh. 20,000. In one month, she sold 20 radios at Sh. 1,500 each and 16 flat iron boxes at Sh. 2,500 each. What was her total earnings for that month?
10. The diagram below represents a solid triangular prism. Calculate its volume.



11. A man deposited Sh. 24,000 in a bank which paid simple interest at the rate of 15% per annum. After some time, he withdrew all his money. If he withdrew Sh. 33,000/=, for how many years had he deposited his money?
12. A bus left Kisumu on Monday at 2225hrs and took 9 hours 45 minutes to reach Nairobi. What day and time did it reach its destination?
13. A truck was loaded with 4.3tonnes of tea leaves which were packed in 100g, 250g, 500g and 1kg packets. The number of 250g packets was three times the number of 500g packets. The number of 500g packets was twice the number of 1kg packets. If the number of 1kg packets was 1000, what was the number of 100g packets?
14. Fruit juice is packed in jerricans each holding 5 litres of juice. Four jerricans are packed in one carton. At a birthday party, 10 cartons of fruit juice were drunk by the invited guests. If each guest drank 400 millilitres of fruit juice, how many guests were there in the party?



15. A contractor employed 60 men in order to build a house in 150 days. How many more days would 50 men take to build the house if they work at the same time?

16. What is the value of $\frac{2}{3}a^2b - \frac{1}{4}abc^2$ when $a=3$, $b=4$ and $c=2$?

17. What is the value of?

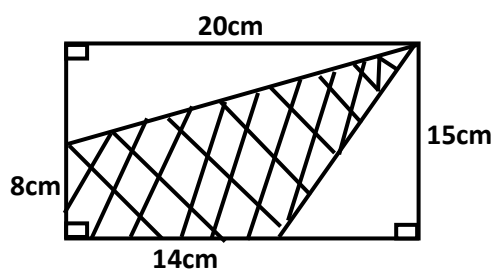
$$\frac{0.364 \times 0.6}{0.0728}$$

18. By selling a jacket for Sh. 1,950 a draper made a profit of 30%. If he had sold the jacket for Sh. 1350, what percentage profit or loss would he have made?

19. What is the value of $1.507 + 3.841 - 0.0564$ correct to 2 decimal places?

20. A man deposited Sh. 30,000 in a bank which paid compound interest at 12% per annum. At the end of the first year he withdrew Sh. 5,600 from his account and left the balance in the account for another year. How much money was in his account at the end of the second year?

21. In the figure below ABCD is a rectangle in which $BC = 15\text{cm}$ and $CD = 20\text{cm}$. BCN and CDM are right angled triangles. M and N are points on AD and AB such that $AM = 8\text{cm}$ and $AN = 14\text{cm}$. What is the area of the shaded part in square centimetres?

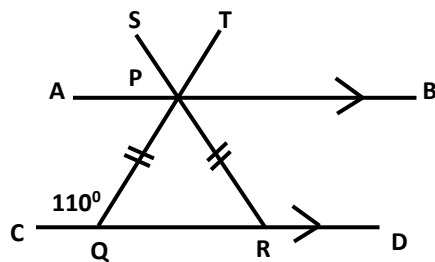




22. The average height of a group of 16 pupils was 125cm. When another 4 pupils joined the group, the new average height is 124cm. what was the average height of the 4 new pupils?

23. What is the value of $\frac{1}{2}$ of $(4\frac{7}{10} - 1\frac{1}{2}) \div \frac{4}{5} + 1\frac{1}{3} + 1\frac{1}{2}$

24. In the figure alongside, line AB is parallel to line CD. PQR is a triangle in which PQ = PR and $\angle PQC = 110^\circ$. QPT and RPS are straight lines. What is the size of angle RPT?



25. The hire purchase price for a wall cabinet is 25% more than the cash price. Kalulu bought the cabinet on hire purchase terms by paying a deposit of Sh. 13,200 and the remaining amount in 12 equal monthly instalments. If the cash price was Sh. 24,000, how much was each monthly instalment?
26. On line AB 6cm, construct triangle ABC in which $BC = CA = 8\text{cm}$. On the triangle construct a circle Centre O which touches the edges of the triangle ABC. What is the radius of the circle?



27. Construct triangle GHT in which line $GH = 5\text{cm}$, $HJ = 7\text{cm}$ and angle $GHJ = 105^\circ$. Construct a circle to pass through the vertices of the triangle. What is the diameter of the circle?

28. Construct triangle XYZ in which $XY = 6\text{cm}$, angle $XYZ = 53^\circ$ and angle $YZX = 62^\circ$. Drop a perpendicular line from vertex X and let it meet line YZ at N. what is the length of line YN?



29. Construct a semi-circle whose diameter $AB = 7.2\text{cm}$. line $AC = 6\text{cm}$ such that C is on the circumference, join C to B to form a triangle. Draw a perpendicular line from C to AB such that the perpendicular line meets line AB at D . what is the length of line DA ?

30. Construct a quadrilateral $PQRS$ in which $PQ = 5.5\text{cm}$, $QR = 7\text{cm}$, $PS = 4\text{cm}$, angle $PQR = 135^\circ$ and angle $QPS = 45^\circ$. Join S to R . drop a perpendicular line from S to meet line PQ at T . What is the size of angle RST ?